

RF-based Dual-mode Robot



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Here is a dual-mode robot that can be operated manually using an RF-based remote control and can also move automatically avoiding all the obstacles. The robot has some inbuilt intelligence to avoid obstacles by changing its path. It can be further modified for applications like automatic vacuum cleaner. In such an application the vacuum cleaner will automatically clean the floor, or you can direct it using the RF remote while sitting relaxed on your sofa.

Circuit and working

Fig. 1 shows the block diagram for PIC-based dual-mode robot. When a button is pressed in the remote, the corresponding code is transmitted through RF transmitter. At the receiver end, this transmitted signal is received and decoded. The MCU unit

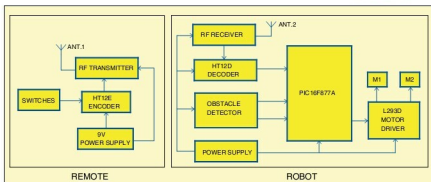


Fig. 1: Block diagram for PIC-based dual-mode robot

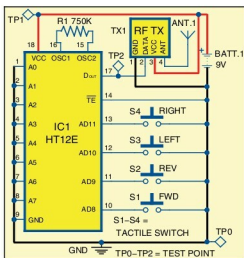


Fig. 2: Circuit of remote control

Test Points of Transmitter

Test point	Details
TP0	Ground (GND)
TP1	9V
TP2	Serial encoded data

compares the received code and drives the motors corresponding to the code received. Obstacle detectors detect any object in the way and intimate the microcontroller.

Figs 2 and 3 show the circuits of remote control and robot, respectively.

Remote control. It is built around encoder IC HT12E (IC1), 433MHz RF transmitter module (TX1) and a few discrete components. Switches S1, S2,

S3 and S4 are interfaced with AD8 through AD11 of encoder IC1 for forward (FWD), reverse (REV), left (LEFT) and right (RIGHT) motions, respectively. Resistor R1 is connected between oscillator pins 15 and 16 to set the transmitter frequency. IC1 is permanently enabled for transmission by connecting its TE pin to ground.

When any switch, say S1, is pressed, the corresponding data is serially transmitted from D_{OUT} pin through the RF transmitter module. A 9V battery is used to power the circuit.

Robot. It comprises an RF receiver module (RX1), decoder HT12D (IC2), microcontroller PIC-16F877A (IC3), operational amplifier LM324 (IC5), motor driver L293D (IC6), regulator 7805 (IC4) and a few discrete components. Data transmitted from the remote is received by the RF receiver module and further decoded by IC2. Decoded data is fed to pins RA0 through RA3 of microcontroller IC3. Identical addresses are selected for the encoder and the decoder by shorting them to ground. Output pins RD0 through RD3 of the microcontroller are fed to IN1 through IN4 of driver IC6 to drive motors M1 and M2, respectively. Enable pins EN1 (pin 1) and EN2 (pin 9) are kept high for always-enabled output.

PARTS LIST

Semiconductors:

IC1	- HT12E encoder
IC2	- HT12D decoder
IC3	- PIC16F877A microcontroller
IC4	- 7805, 5V regulator
IC5	- LM324 operational amplifier
IC6	- L293D motor driver
IRTX1, IRTX2	- IR transmitter
IRRX1, IRRX2	- IR receiver
LED1-LED4	- 5mm LED
D1	- 1N4007 rectifier diode
TX1	- ASK transmitter module
RX1	- ASK receiver module

Resistors (all 1/4-watt, ±5% carbon):

R1	- 750-kilo-ohm
R2	- 33-kilo-ohm
R3, R8, R9	- 10-kilo-ohm
R4, R5, R10, R11	- 330-ohm
R6, R7	- 100-ohm
VR1, VR2	- 10-kilo-ohm preset

Capacitors:

C1, C2, C3	- 0.1µF ceramic disk
C4, C5	- 22µF ceramic disk

Miscellaneous:

S1-S4, S6	- Push-to-on tactile switch
S5	- On/off switch
M1, M2	- 12V, 100-rpm geared motor
BATT.1	- 9V battery (TP3)
BATT.2	- 12V, 4.5Ah battery
X _{TAL1}	- 4MHz crystal